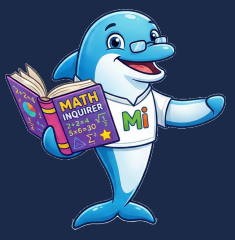
TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 3 OF 10

# Digit Cards




**Big idea:** The order of the digits changes the number. Two digit cards make two different numbers. The left card counts the tens, so the card on the left matters most.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

## 1. At a glance

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|---|---|
| **Strand** | Whole Numbers (Numbers to 100: compare and order numbers; Compare two sets of objects; Tens and ones (place value) used to compare)<br>Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as 'MOE-aligned'. Comparing and ordering numbers to 100 and comparing sets of objects are Primary 1 topics. Term 1 goes beyond the school Primary 1 syllabus on purpose: the Go further puzzles use four cards and several clues. This lesson stays within two-digit numbers; the one beyond-syllabus challenge item is G2 part c (how many different 3-digit numbers can 3 cards make, each used once: 6), a Go further listing item; no other three-digit numbers are used in the workbook. |
| **Objectives** | • Make two-digit numbers from two digit cards, and know that swapping the cards makes a different number.<br>• Compare two numbers by looking at the tens first, then at the ones, and say which is greater.<br>• Make the greatest and the smallest number from two or three cards, including when one card is 0.<br>• Put up to four two-digit numbers in order, smallest first. |
| **Joining cold** | **Remember warm-up covers:** A digit is one of the symbols 0 to 9; Tens and ones: 3 bars and 4 squares is 34 (the left digit counts the tens); More or fewer: which group has more (squares or counters, to 20). Only counting to 20 and the idea of more and fewer. Tens and ones is taught again with the kit bars and squares in the warm-up, so a child who missed the first lessons of the term can join. Nothing from earlier lessons is assumed except through the warm-up.<br>Not one of the hardest to miss (plan, Term 1 note). A child who misses it watches the recording, makes the digit cards, and does workbook pages 1 and 2 before Lesson 4. Comparing 2-digit numbers comes back in Lessons 5 to 7, and each of those warm-ups teaches it again. |
| **Child materials** | • Digit cards: cut out the two sets of 0 to 9 from the kit page, or write 0 to 9 twice on 20 small scraps of paper (a grown-up helps, before the lesson)<br>• The printed workbook (pages 1 to 5) and the two kit pages (tens bars and ones squares, and digit cards), already cut out by a grown-up<br>• A pencil, a rubber and spare paper |
| **Tutor materials** | • A slide deck and your own set of digit cards 0 to 9 to show on camera if you wish<br>• A timer slide or a visible clock<br>No ruler is needed. Nothing needs to be bought: scrap paper and a pencil are enough. No matches, toothpicks or anything sharp. If scissors are used to cut the paper, a grown-up cuts. |
| **Tiers** | Start here (8 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit pages (pages 6-7). Labels are neutral; nobody is told which page they are on. |
| **Contest-style move** | Used at Think harder only: digit-card clue puzzle (find every number that fits two clues); list all the cases systematically (every number from the cards); a number cannot start with 0 (smallest and greatest number with a 0 card). Never advertised to children or parents. |

**Before the lesson**

- Solve every workbook question yourself first, including the lists in Think harder and Go further.
- Tell parents a week before: print the whole workbook (the five lesson pages and the two kit pages), and cut out both kit pages (digit cards, and tens bars and ones squares). If a family cannot print, 20 scraps of paper (0 to 9 written twice, by a grown-up before the lesson) work as digit cards.
- Make your own set of digit cards 0 to 9 to show on camera. Have the slides open and check the timer slide and the kit page are visible on screen share.
- Test the recording once with several tiles before the first live class: check that it captures the tutor, the shared screen and any voice that is not muted. Write the setting you used in your tutor notes. Say the recording statement in the same words every lesson.
- Tell families plainly that Term 1 goes beyond the school Primary 1 syllabus on purpose. Start here is approachable with the kit and is explored in class; Go further is optional and not a claim about school exams.

**Safeguarding and recording (this lesson)**

- Cards are paper: nothing sharp, and no scissors in the lesson (a grown-up cuts the kit pages beforehand). Small pieces stay on the table and out of mouths and ears. Keep small things away from younger brothers and sisters.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for up to 90 days. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

## 2. Timing plan (90 minutes exactly)

| Minutes | Part | Min | Slides | Script fills |
|---|---|---|---|---|
| 0–3 | **Opening** | 3 | 2: Digit Cards, Get ready | 2:38 |
| 3–9 | **Remember warm-up** | 6 | 3: Warm-up: count in tens, Tens and ones, More or fewer? | 4:54 |
| 9–13 | **Sneak-peek puzzles** | 4 | 2: Sneak-peek puzzle 1, Sneak-peek puzzle 2 | 3:29 |
| 13–28 | **Explore** | 15 | 4: Explore: two cards, two numbers, Build both numbers, Show me!, What did we notice? | 13:29 |
| 28–38 | **Mini-lesson** | 10 | 7: Two cards, two numbers, Which is greater? Tens first, Your turn: which is greater?, Check: which is greater?, Greatest and smallest, The 0 card, Put them in order | 8:07 |
| 38–42 | **Brain break** | 4 | 1: Tall and small | 3:24 |
| 42–70 | **Workbook** | 28 | 5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page | 26:53 |
| 70–76 | **Talk time** | 6 | 2: Talk and explain, Back to the sneak-peek | 5:30 |
| 76–82 | **Exit ticket** | 6 | 2: Exit ticket, Check the exit ticket | 4:07 |
| 82–85 | **Look-back** | 3 | 1: Look-back | 2:43 |
| 85–90 | **Flex time and goodbye** | 5 | 1: Well done, inquirers! | 0:44 |

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| | **Total** | **90** | | **75:58** |

The script needs 75:58 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 89:57. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 13 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

## 3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

### 0:00 – 3:00 Opening

script 2:38 of 3:00

| | | |
|---|---|---|
| 0:00 | SLIDE | Show slide **Digit Cards** (s-title) |
| 0:25 | DO | Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12] |
| 0:37 | SAY | Hello, inquirers! Your digit cards are out. Cameras can be on or off, as you like. |
| 0:45 | SAY | This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off. |
| 1:02 | DO | Start the recording now. Do not use children's first names on the recording. [0:05] |
| 1:08 | REC | *If you are watching the recording: hello! Make your digit cards 0 to 9, get your workbook and a pencil. Pause or rewind any time.* |
| 1:20 | SAY | Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera. |
| 1:35 | SLIDE | Show slide **Get ready** (s-cards) |
| 2:00 | READ | Check your things. Your digit cards, from 0 to 9, face up. Your tens bars and ones squares. A pencil and your workbook. |
| 2:13 | DO | Greet the whole group warmly. Do not use children's first names on the recording. A child whose digit cards are not cut out can still join the warm-up, which needs no cards; a grown-up helps them write 0 to 9 twice on 20 scraps. [0:25] |
| | NOTE | The routine and the four inquirer rules were taught in Lesson 1. Name them only if someone has joined who missed it: notice it, test it, explain why, mistakes help us. Say the recording notice in your own words: the recording has the tutor, the shared screen and any unmuted voice. |
| | NOTE | New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. The first recording of the term shows the full routine. |

### 3:00 – 9:00 Remember warm-up

script 4:54 of 6:00

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|---|---|---|
| 3:00 | SLIDE | Show slide **Warm-up: count in tens** (s-rem1) |
| 3:25 | REC | *Recording: join in! Say the numbers out loud and show tens and ones with your bars and squares. Pause if you need longer.* |
| 3:36 | SAY | Stand up if you can. We clap on every number. Count in tens with me: ten, twenty, thirty, all the way to one hundred. |
| 3:48 | CHILDREN | Count in tens together, clapping, to 100. Then back from 100 to 0, whispering, shouting on 50. [0:40] |
| 4:28 | SLIDE | Show slide **Tens and ones** (s-rem2) |
| 4:54 | SAY | Now tens and ones. A digit is one of these number symbols: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Look at 34. The left digit counts the tens bars. The right digit counts the ones squares. Three bars and four squares make thirty-four. |
| 5:16 | ASK | Twenty-five. How many tens? How many ones? [wait about 14 s for answers] |
| 5:34 | ASK | Forty. How many tens? How many ones? [wait about 14 s for answers] |
| 5:52 | ASK | Seventeen? [wait about 14 s for answers] |

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|---|---|---|
| 6:06 | **ASK** | Sixty-three? [wait about 14 s for answers] |
| 6:20 | **SLIDE** | Show slide **More or fewer?** (s-rem3) |
| 6:46 | **SAY** | More or fewer? Look at the two groups of squares. Match them up one to one if you like. |
| 6:55 | **ASK** | Which group has more squares, group A or group B? [wait about 12 s for answers] |
| 7:12 | **SAY** | Group A is a full ten and three more: thirteen. Group B is nine, less than a full ten. So group A has more. |
| 7:24 | **ASK** | Sixteen or nineteen: which is more? [wait about 12 s for answers] |
| 7:39 | **ASK** | Twenty or twelve: which is more? [wait about 12 s for answers] |
| | **IF STUCK** | If a child hesitates on twenty or twelve, ask them to build both with bars and squares: 2 bars is more than 1 bar and 2 squares. Say the tens word slowly and point to the tens bars. |
| | **NOTE** | Answers: 2 tens 5 ones, 4 tens 0 ones, 1 ten 7 ones, 6 tens 3 ones; group A (13) has more than group B (9); 19; 20. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: counting to 20 is enough, because tens and ones are shown with the bars and squares. |

## 9:00 – 13:00 Sneak-peek puzzles

script 3:29 of 4:00

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|---|---|---|
| 9:00 | **SLIDE** | Show slide **Sneak-peek puzzle 1** (s-sneak) |
| 9:25 | **REC** | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* |
| 9:34 | **SAY** | Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked. |
| 9:47 | **READ** | Puzzle one. Count the stars in box A and box B. Which box has more stars? Write how many stars are in that box. |
| 10:00 | **CHILDREN** | Children count the stars and write the number on their own paper. [0:45] |
| 10:45 | **SLIDE** | Show slide **Sneak-peek puzzle 2** (s-sneak2) |
| 11:10 | **READ** | Puzzle two. Ali has the digit cards 4 and 8. He puts the 4 first and makes 48. Mei swaps the cards and makes a new number. Who has the greater number? Circle Ali or Mei. I know because ___. |
| 11:31 | **CHILDREN** | Children write or draw their answer. [0:50] |
| 12:21 | **SAY** | Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time. |
| | **NOTE** | Puzzle 1 answer: box B has more, 18 stars (box A has 15). Puzzle 2 answer: Mei, because 84 is 8 tens and 48 is 4 tens. Look for good ideas to praise later (a child who built or drew the two numbers, or who said 8 tens is more than 4 tens). Do not write scores or keep a list. |

## 13:00 – 28:00 Explore

script 13:29 of 15:00

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| 13:00 | **SLIDE** | Show slide **Explore: two cards, two numbers** (s-explore1) |
| 13:25 | **REC** | *Recording: pause now. Pick two cards, make both numbers, then build them with bars and squares. Press play when you have tried.* |
| 13:36 | **SAY** | Today your cards are the teachers. Pick any two cards and put them side by side. Read your number, tens first. Now swap the cards and read the new number. Is it the same? |
| 13:53 | **SAY** | Try three different pairs. If you finish, try the what-if card: what if one card is 0? |
| 14:02 | **CHILDREN** | Round 1: each child makes and swaps three pairs of cards and says the numbers aloud. Tutor scans the cameras. Children keep their cards on the table and say both numbers aloud when asked; nobody holds anything up to the camera. [2:30] |
| 16:32 | **ASK** | Tell me what you noticed when you swapped the cards. Was the number the same? [wait about 45 s for answers] |
| 17:24 | **SLIDE** | Show slide **Build both numbers** (s-explore2) |
| 17:49 | **SAY** | Now check with your bars and squares. Pick two cards that add up to 9 or less, like 3 and 5, so you have enough bars. Build both numbers. Bars in a row for the tens, squares for the ones. |

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| 18:09 | **CHILDREN** | Children build both numbers with the kit; a child may build one number, read it, then build the other. Tutor scans the cameras; showing is for volunteers. [1:50] |
| 19:59 | **ASK** | Which pile has more? Which number is greater? How do you know? [wait about 45 s for answers] |
| 20:50 | **SAY** | The pile with more bars has the greater number. That is why we look at the tens first. |
| 20:59 | **SLIDE** | Show slide **Show me!** (s-explore3) |
| 21:24 | **SAY** | Show me! I say two cards. You make the greatest number you can. Give a thumbs-up when it is made, then check your own against my line. |
| 21:38 | **SAY** | Cards seven and two. Greatest number. |
| 21:40 | **CHILDREN** | Children make 72 and give a thumbs-up. [0:30] |
| 22:10 | **READ** | Check your own: seven on the left, two on the right. Seventy-two. |
| 22:17 | **SAY** | Cards three and nine. |
| 22:19 | **CHILDREN** | Make 93. [0:30] |
| 22:49 | **READ** | Check your own: nine on the left, three on the right. Ninety-three. |
| 22:56 | **SAY** | Cards zero and five. Is there only one number you can make? Think about it. |
| 23:03 | **CHILDREN** | Children try 50 and 05. Do not settle the zero yet: leave it for the next slide. [0:40] |
| 23:43 | **SAY** | Now the smallest number. Cards six and one. |
| 23:47 | **CHILDREN** | Make 16. [0:30] |
| 24:17 | **READ** | Check your own: one on the left, six on the right. Sixteen. |
| 24:24 | **SLIDE** | Show slide **What did we notice?** (s-explore4) |
| 24:49 | **ASK** | Which card went on the left to make the greatest number? And for the smallest number? [wait about 40 s for answers] |
| 25:37 | **ASK** | What happened with the 0 card and the 5 card? Could you make 05? [wait about 30 s for answers] |
| 26:14 | **SAY** | Greatest number: put the greatest card on the left. Smallest number: put the smallest card on the left. But a number does not start with 0, so 05 is just 5. |
| | **IF STUCK** | A child who reads 47 as seven-four: ask them to build it with 4 bars and 7 squares and say the tens first. |
| | **IF STUCK** | A child with no cards: use the digits from the kit, or write them on scraps of paper while the others do round 1. |
| | **NOTE** | Main room only: no breakout rooms, so you can see every camera. Pairs are optional: a grown-up may check the numbers. Everyone builds with the kit; the what-if card is open to anyone and nobody is named. |

## 28:00 – 38:00 Mini-lesson

script 8:07 of 10:00

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| 28:00 | **SLIDE** | Show slide **Two cards, two numbers** (s-m1) |
| 28:25 | **REC** | *Recording: keep watching and copy the examples. Rewind if you miss a step.* |
| 28:32 | **SAY** | Look. The cards 3 and 5. Three first is thirty-five. Five first is fifty-three. Same cards, two different numbers. The left card counts the tens. |
| 28:44 | **ASK** | Which card is the tens card in fifty-three? [wait about 12 s for answers] |
| 29:00 | **SLIDE** | Show slide **Which is greater? Tens first** (s-m2) |
| 29:25 | **SAY** | Which is greater, 61 or 59? Look at the bars. 6 bars is more than 5 bars. So 61 is greater, even though 9 is a greater digit than 1. We look at the tens first. |
| 29:43 | **SAY** | If the tens are the same, we look at the ones. Forty-seven and forty-three have 4 tens each. 7 ones is more than 3 ones. |

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| 29:56 | **ASK** | Which is greater, 52 or 58? How do you know? [wait about 20 s for answers] |
| 30:20 | **SLIDE** | Show slide **Your turn: which is greater?** (s-m3) |
| 30:46 | **READ** | Your turn. Write the greater number each time. 28 or 82. 61 or 59. 34 or 38. |
| 30:55 | **CHILDREN** | Children write three numbers. [1:00] |
| 31:55 | **SLIDE** | Show slide **Check: which is greater?** (s-m3a) |
| 32:20 | **READ** | Eighty-two. Sixty-one. Thirty-eight. |
| 32:21 | **SLIDE** | Show slide **Greatest and smallest** (s-m4) |
| 32:46 | **SAY** | Cards 7 and 2. For the greatest number, put the greatest card on the left: seventy-two. For the smallest, put the smallest card on the left: twenty-seven. |
| 33:00 | **ASK** | Cards 4 and 9. What are the greatest and the smallest numbers? [wait about 25 s for answers] |
| 33:31 | **SAY** | Ninety-four and forty-nine. |
| 33:32 | **ASK** | Cards 1 and 6. Greatest number? Smallest number? [wait about 25 s for answers] |
| 34:01 | **SAY** | Sixty-one and sixteen. |
| 34:03 | **SLIDE** | Show slide **The 0 card** (s-m5) |
| 34:28 | **SAY** | The 0 card is tricky. Cards 0 and 6. Six then zero is sixty. Zero then six is 06, and that is just 6. We do not start a number with 0, so sixty is the only two-digit number. |
| 34:47 | **ASK** | Why is 06 not a two-digit number? [wait about 15 s for answers] |
| 35:06 | **SLIDE** | Show slide **Put them in order** (s-m6) |
| 35:31 | **READ** | Arrange these in order. Begin with the smallest. 45, 54, 39, 40. Look at the tens first. |
| 35:40 | **CHILDREN** | Children say or write the order. [0:25] |
| 36:05 | **SAY** | Thirty-nine, forty, forty-five, fifty-four. |
| | **NOTE** | Key words to use every time: digit, tens digit, ones digit, greater, smaller. Do not teach the greater-than and less-than symbols. Do not teach three-digit numbers; this lesson stays within two-digit numbers. Keep the zero rule short: 0 can not go on the left. |
| | **IF STUCK** | If a child says 59 is greater than 61 because 9 is greater than 1: ask them to build both with bars and squares and count the bars first. |

## 38:00 – 42:00 Brain break

script 3:24 of 4:00

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| 38:00 | **SLIDE** | Show slide **Tall and small** (s-break) |
| 38:25 | **REC** | *Recording: stand up and join in! Reach up tall for the greater number, curl small for the smaller one.* |
| 38:34 | **SAY** | Stand up and get a sip of water. I say two numbers. If the first one is greater, reach up tall. If the first one is smaller, curl down small. If they are the same, hands on hips. |
| 38:54 | **CHILDREN** | Call: 52 and 25 (tall), 38 and 83 (small), 61 and 59 (tall), 40 and 40 (hands on hips), 19 and 21 (small). Pause after each so children can say how they know. [2:30] |

## 42:00 – 70:00 Workbook

script 26:53 of 28:00

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| 42:00 | **SLIDE** | Show slide **Workbook page 1: Start here** (s-ws1) |
| 42:25 | **REC** | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* |
| 42:34 | **READ** | Page 1, Start here. Question 1: make two numbers with the cards 3 and 5. Question 2: write the greater number each time. Question 3: look at the bars and squares and say who has more. |
| 42:54 | **CHILDREN** | Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:00] |
| 50:54 | **SLIDE** | Show slide **Check page 1** (s-ws1ans) |

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|---|---|---|
| 51:19 | **READ** | Thirty-five and fifty-three. Seventy-two, forty-eight, sixty-five, thirty. Mei has thirty-eight, Ben has forty-one, Ben has more. Mark your own page in another colour. |
| 51:31 | **SAY** | After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time. |
| 51:40 | **SLIDE** | Show slide **Workbook page 2: Start here** (s-ws2) |
| 52:05 | **READ** | Page 2. Question 4: make two numbers with the cards 6 and 2. Question 5: arrange four numbers in order. Question 6: Kumar says that 19 is greater than 21. Is Kumar right? Question 7: the cards 2 and 8. Question 8: Siti and Ravi have stickers, so draw bars and squares. |
| 52:34 | **CHILDREN** | Page 2, questions 4 to 8. [10:00] |
| 62:34 | **SLIDE** | Show slide **Check page 2** (s-ws2ans) |
| 62:59 | **READ** | Sixty-two is greater and twenty-six is smaller. Fourteen, twenty-five, forty-one, fifty-two. Kumar is not right: twenty-one is greater. Eighty-two and twenty-eight. Ravi has more stickers. |
| 63:12 | **SLIDE** | Show slide **Choose your next page** (s-choice) |
| 63:37 | **SAY** | When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too. |
| 63:53 | **CHILDREN** | Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [5:00] |
| | **IF STUCK** | Q1: let the child put the two cards down and read them, tens first. Q3: ask how many bars each child has. Q5: ask which numbers have the fewest tens. Q6: build both numbers with the kit and count the bars. Q7: try putting the 8 on the left, then the 2. Q8: draw a bar as a long rectangle and a square as a small box. |
| | **IF STUCK** | A child who is still stuck ticks 'I am still stuck' on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp. You then help in the open session in the next lesson. Never a private message to a child. |
| | **NOTE** | Think harder (page 3) is for the same idea in a new setting: three cards, the 0 card, clue puzzles and listing every number. Go further (page 4) has a four-card puzzle, counting every number and a three-clue puzzle. Do not announce who is on which page. Pairs are never split unless both children agree. Slow class: the workbook part needs about 32:00 against 28:00, so Start here may fill most of it. Take the extra minutes from the flex time, do not skip the answer slides, and say that pages 3 to 5 are for home. |

## 70:00 – 76:00 Talk time

script 5:30 of 6:00

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| 70:00 | **SLIDE** | Show slide **Talk and explain** (s-talk) |
| 70:25 | **REC** | *Recording: pause. Tell a grown-up your answer to the talk question, and say why.* |
| 70:32 | **READ** | Put the smallest card on the left to make the smallest number. Is this always right? Part a: try the cards 3 and 8. Does it work? Part b: try the cards 0 and 5. Does it work? Circle Yes or No for each. I know because ___. |
| 70:58 | **CHILDREN** | Children try both pairs of cards and say what they find. [0:50] |
| 71:48 | **CHILDREN** | Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00] |
| 72:48 | **ASK** | Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers] |
| 74:24 | **SLIDE** | Show slide **Back to the sneak-peek** (s-sneakans) |
| 74:49 | **SAY** | Back to the sneak-peek. Box B had more stars. For puzzle two, Mei has the greater number: eighty-four is 8 tens, and Ali has forty-eight, only 4 tens. |
| 75:03 | **ASK** | Did anyone change their mind about a puzzle while we worked? You can just nod. [wait about 20 s for answers] |

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| | NOTE | Model answer: (a) yes, 38 is smaller than 83. (b) no: 0 can not go on the left because 05 is just 5, so 50 is the only two-digit number. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed. |

## 76:00 – 82:00 Exit ticket

script 4:07 of 6:00

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|---|---|---|
| 76:00 | SLIDE | Show slide **Exit ticket** (s-exit) |
| 76:25 | REC | *Recording: pause and do the exit ticket on your own. Then press play to check it.* |
| 76:33 | SAY | Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5. |
| 76:41 | READ | E1. Write the greater number: thirty-nine or forty-one. E2. Make two numbers with the cards 7 and 3. Write the greater and the smaller. E3. Arrange in order, smallest first: sixty-three, thirty-six, sixty. |
| 76:59 | CHILDREN | Children answer E1 to E3 on their own. [2:30] |
| 79:29 | SLIDE | Show slide **Check the exit ticket** (s-exitans) |
| 79:54 | READ | Forty-one. Seventy-three and thirty-seven. Thirty-six, sixty, sixty-three. |
| 79:58 | SAY | Check your own ticket in a different colour. Well done. Put a star by each one you got right. |
| | NOTE | Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child's result. |

## 82:00 – 85:00 Look-back

script 2:43 of 3:00

| | | |
|---|---|---|
| 82:00 | SLIDE | Show slide **Look-back** (s-look) |
| 82:25 | REC | *Recording: pause and fill in your look-back box. You can pass.* |
| 82:30 | SAY | Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick 'I am still stuck', ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you. |
| 83:03 | CHILDREN | Children fill in the look-back box on page 5. [1:40] |

## 85:00 – 90:00 Flex time and goodbye

script 0:44 of 5:00

| | | |
|---|---|---|
| 85:00 | SLIDE | Show slide **Well done, inquirers!** (s-end) |
| 85:25 | SAY | Well done, inquirers. Next lesson is The 100 Square Explorer. Please have the printed 100 square page, about 12 small counters, two or three colours, and a pencil and paper. Goodbye, and thank you to the grown-ups. |
| | SLIDE | Show slide **Go further: what if?** (s-opt) *(optional)* |
| | SAY | Optional, only if you are on time: how many two-digit numbers have two matching digits, like 33? Write them all. |
| | CHILDREN | Short think time, then take two answers. [1:30] |
| | SLIDE | Show slide **If we have time: pick one** (s-menu) *(optional)* |
| | SAY | Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each). |
| | NOTE | This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 11, 22, 33, 44, 55, 66, 77, 88, 99, nine numbers, each one 11 more than the one before (do not teach adding 11 here). If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on. |

## 3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

| Activity | Minutes | You say | Recording cue | Answer |
|---|---|---|---|---|
| **Is it always true?** | 4 (optional) | Mei says: the greater card on the left always makes the greater number. Is that always true? Try the pairs 2 and 9, 5 and 6, and 3 and 7 with your cards. | *Recording: pause. Try three pairs of cards. Is the greater card on the left always the greater number?* | Yes, for two different cards. The left card counts the tens, so it decides. (If one card is 0 there is only one two-digit number, with the other card on the left.) |
| **Make a puzzle for your grown-up** | 4 (optional) | Pick three digit cards. Write two clues, like: it is more than 30, and the right digit is greater than the left digit. Give it to your grown-up to solve. | *Recording: pause. Pick three cards and write a two-clue puzzle for a grown-up to solve.* | No single answer. Check that the clues fit the secret number and say how many numbers fit. |
| **Teach it back** | 3 (optional) | Pretend your grown-up or a toy has never compared numbers. Teach them how to tell that 61 is greater than 59, using tens and ones. | *Recording: pause. Teach a grown-up how to tell that 61 is greater than 59.* | 61 has 6 tens and 59 has 5 tens, so 61 is greater. We look at the tens first. The ones only matter when the tens are the same. |

# 4. Questions, answers, hints and common mistakes

## Sneak-peek puzzles (minutes 9 to 13)

**Puzzle 1 (access):** Count the stars in box A and box B. Which box has more stars? Write how many stars are in that box.

> **Answer: 18**

**Puzzle 2 (think):** Ali has the digit cards 4 and 8. He puts the 4 first and makes 48. Mei swaps the cards and makes a new number. Who has the greater number? Circle Ali or Mei. I know because ___.

> **Answer: Mei. Her number is 84, which is 8 tens. Ali has 48, which is only 4 tens.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

## Start here (workbook pages 1 and 2)

**S1** **You have two digit cards: 3 and 5. Make two numbers with both cards.**

(a) 3 first, then 5: ___
(b) 5 first, then 3: ___
(c) The greater number is ___

> **Answer: 35 and 53. 53 is greater.**

**Hint (say this first):** Put the cards down side by side. Say the tens first. Swap them and say the new number.
**Worked solution:** With 3 on the left the number is 3 tens and 5 ones: 35. With 5 on the left it is 5 tens and 3 ones: 53. The same two cards make two numbers. 53 is greater, because 5 tens is more than 3 tens.
**Common mistakes:** Writing 8 (3 + 5).; Writing 35 twice.
Time for a middle child: about 1.5 min.

**S2** **Write the greater number each time.**

(a) 27 or 72: ___
(b) 48 or 41: ___
(c) 65 or 56: ___
(d) 13 or 30: ___

> **Answer: 72, 48, 65, 30**

**Hint (say this first):** Look at the tens first. If the tens are the same, look at the ones. You can build both with your kit.
**Worked solution:** (a) 7 tens is more than 2 tens: 72. (b) Same tens, 8 ones is more than 1 one: 48. (c) 6 tens is more than 5 tens: 65. (d) 3 tens is more than 1 ten: 30.
**Common mistakes:** Choosing the number with the greater ones digit (saying 13 is greater than 30).; Not looking at the tens first.
Time for a middle child: about 3 min.

**S3** **Count the bars and squares in Mei's picture. Ben has 4 bars and 1 square.**



(a) Mei's number is ___
(b) Ben's number is ___
(c) ___ has the greater number.

> **Answer: Mei 38, Ben 41, Ben has more**

**Hint (say this first):** Count the bars for the tens and the squares for the ones. Then look at the tens first.
**Worked solution:** Mei has 3 bars and 8 squares: 38. Ben has 4 bars and 1 square: 41. 4 tens is more than 3 tens, so Ben has the greater number, even though Mei has more squares.
**Common mistakes:** Saying Mei has more because she has 8 squares.; Reading Mei's picture as 83.
Time for a middle child: about 2.5 min.

**S4** **Make two numbers with both cards 6 and 2.**

(a) The greater number is ___

(b) The smaller number is ___

**Answer: 62 and 26**

**Hint (say this first):** Put the cards down both ways and read them. The greater card on the left makes the greater number.

**Worked solution:** The greater card on the left makes the greater number: 62. The smaller card on the left makes the smaller number: 26.

**Common mistakes:** Writing 62 for both.; Writing 8 (6 + 2).

Time for a middle child: about 1.5 min.

**S5** **Arrange these in order. Begin with the smallest number: 52, 25, 41, 14.**

___ , ___ , ___ , ___

**Answer: 14, 25, 41, 52**

**Hint (say this first):** Look at the tens first. Which number has the fewest tens?

**Worked solution:** 14 has 1 ten, 25 has 2 tens, 41 has 4 tens, 52 has 5 tens. Smallest first: 14, 25, 41, 52.

**Common mistakes:** Putting 25 after 41 because 5 is greater than 1.; Writing them greatest first.

Time for a middle child: about 2 min.

**S6** **Kumar says that 19 is greater than 21.**

(a) Is Kumar right? Circle Yes or No.

(b) The greater number is ___

**Answer: No. 21 is greater. 2 tens is more than 1 ten.**

**Hint (say this first):** Build both with your kit. Count the bars first.

**Worked solution:** 19 is 1 ten and 9 ones. 21 is 2 tens and 1 one. 2 tens is more than 1 ten, so 21 is greater. Kumar looked at the ones digits first. Look at the tens first.

**Common mistakes:** Saying Kumar is right because 9 is more than 2.; Not giving a reason (any clear sentence about the tens is fine).

Time for a middle child: about 2.5 min.

**S7** **You have the cards 2 and 8. Use both cards each time.**

(a) Make a number greater than 50: ___

(b) Make a number smaller than 50: ___

**Answer: 82 and 28**

**Hint (say this first):** Try the 8 on the left, then the 2 on the left. Which number is greater than 50?

**Worked solution:** 82 has 8 tens, which is more than 5 tens. 28 has only 2 tens, which is less than 5 tens.

**Common mistakes:** Writing 82 for both parts.; Writing 8 and 2.

Time for a middle child: about 2 min.

**S8** **Siti has 25 stickers. Ravi has 52 stickers. Draw bars and squares in the box.**

___ has more stickers.

**Answer: Ravi (52 is 5 tens and 2 ones; 25 is 2 tens and 5 ones)**

**Hint (say this first):** A bar is a ten and a square is a one. Draw the bars first, then the squares.

**Worked solution:** Siti: 2 bars and 5 squares. Ravi: 5 bars and 2 squares. 5 bars is more than 2 bars, so Ravi has more stickers, even though both numbers use a 2 and a 5.

**Common mistakes:** Saying they have the same because they use the same digits.; Drawing 2 and 5 single squares for Siti.

Time for a middle child: about 2.5 min.

## Think harder (page 3): the taught idea in a new setting

**T1** **You have the cards 3, 5 and 8. Use two different cards to make the greatest number. Then make the smallest number.**

(a) The greatest number is ___
(b) The smallest number is ___

**Answer: 85 and 35**

**Hint (say this first):** Pick the greatest card for the tens to make the greatest number. Pick the smallest card for the tens to make the smallest.
**Worked solution:** Numbers you can make: 35, 38, 53, 58, 83, 85. The greatest is 85 (8 tens, the greatest card on the left, then the next greatest card). The smallest is 35.
**Common mistakes:** Using all three cards.; Writing 88 (using the 8 twice).
Time for a middle child: about 4 min.

**T2** **You have the cards 0, 3 and 7. Use two cards to make the smallest and the greatest two-digit numbers.**

(a) The smallest number is ___
(b) The greatest number is ___

**Answer: 30 and 73**

**Hint (say this first):** List the numbers you can make. Can 0 go on the left?
**Worked solution:** The 0 can not go on the left (03 is just 3). The two-digit numbers are 30, 37, 70 and 73. The smallest is 30 and the greatest is 73.
**Common mistakes:** Writing 03 or 3 for the smallest.; Writing 37 for the smallest.
Time for a middle child: about 4 min.

**T3** **You have the cards 1, 3, 5 and 7. Make two-digit numbers with two different cards. The number is more than 40. Its left digit is greater than its right digit. Write all the numbers that fit.**

**Answer: 51, 53, 71, 73, 75**

**Hint (say this first):** First list the numbers that are more than 40. Then cross out any where the left digit is not greater.
**Worked solution:** More than 40 means a tens card of 5 or 7. With 5 on the left: 51, 53, 57. With 7 on the left: 71, 73, 75. Clue 2 removes 57, because 5 is not greater than 7. Left: 51, 53, 71, 73, 75.
**Common mistakes:** Stopping after one or two numbers.; Writing 57 (5 is less than 7).
Time for a middle child: about 5 min.

**T4** **You have the cards 2, 4 and 6. Use two different cards each time. Write all the two-digit numbers you can make.**

There are ___ numbers.

**Answer: 6 numbers: 24, 26, 42, 46, 62, 64**

**Hint (say this first):** Start with 2 on the left, then 4, then 6. Take your time and go in order.
**Worked solution:** 2 on the left: 24, 26. 4 on the left: 42, 46. 6 on the left: 62, 64. That is 2 + 2 + 2 = 6 numbers.
**Common mistakes:** Missing some by listing in a random order.; Writing 22, 44 or 66 (each card can be used only once).
Time for a middle child: about 5 min.

## Go further (page 4): mastery and extension

**G1** **You have the cards 2, 5, 6 and 9. Use two different cards to make a two-digit number.**

(a) The greatest number less than 60 is ___
(b) The smallest number greater than 50 is ___

**Answer: 59 and 52**

**Hint (say this first):** List every two-digit number you can make. Then cross out the numbers that are too big or too small.
**Worked solution:** The numbers are 25, 26, 29, 52, 56, 59, 62, 65, 69, 92, 95, 96. Less than 60: 25, 26, 29, 52, 56, 59, so the greatest is 59. Greater than 50: 52, 56, 59, 62, 65, 69, 92, 95, 96, so the smallest is 52.
**Common mistakes:** Writing 96 or 69 for part a (they are greater than 60).; Writing 25 for part b (it is not greater than 50).
Time for a middle child: about 5 min.

**G2** **You have the cards 1, 2, 3 and 4. Make two-digit numbers. How many numbers can you make?**

(a) Each card used once: ___ numbers
(b) A card used twice is allowed, like 11: ___ numbers
(c) The cards 1, 2 and 3 make 3-digit numbers, each card once: ___ numbers

**Answer: 12, 16 and 6**

**Hint (say this first):** Think about the tens card first. How many choices are there? Then think about the ones card. For part c, list them in order: 1 first, then 2 first, then 3 first.
**Worked solution:** Each card once: 4 choices for the tens card, then 3 choices for the ones card, so 4 x 3 = 12. A card may be used twice: 4 choices and 4 choices, so 16 (the 12 and the four numbers 11, 22, 33, 44). Part c is beyond what school teaches: 3 cards in a row. 1 first: 123, 132. 2 first: 213, 231. 3 first: 312, 321. That is 3 x 2 x 1 = 6 numbers.
**Common mistakes:** Counting only 8 or 6 numbers.; Forgetting 11, 22, 33 and 44 in part b.; Part c: listing in a random order and missing one (there are 6, two for each first card).
Time for a middle child: about 8 min.

**G3** **I make a two-digit number with two different cards from 1, 3, 4 and 7. My number is between 35 and 45. My ones digit is greater than my tens digit. What number am I?**

I am ___

**Answer: 37**

**Hint (say this first):** List all the two-digit numbers you can make with the cards. Cross out the numbers that break each clue in turn.
**Worked solution:** The numbers are 13, 14, 17, 31, 34, 37, 41, 43, 47, 71, 73, 74. Between 35 and 45: 37, 41, 43. Ones digit greater than tens digit: 37 (7 is greater than 3). 41 and 43 have a ones digit that is smaller.
**Common mistakes:** Stopping at 41 or 43 and not checking the ones digit.; Writing 34 or 47 (they are not between 35 and 45).
Time for a middle child: about 6 min.

## Investigation (page 5): Is the greater card always first?

**Easy first step:** Take the cards 2 and 7. Make both numbers. Which number is greater? Which card is on the left?
**Then:** Try the cards 3 and 9, 6 and 5, 0 and 8, 4 and 4. Is the greater card on the left always the greater number? When can you not make two numbers? What do you notice?

**Model answer: For two different cards, the greater card on the left always makes the greater number, because the tens decide. With the cards 0 and 8 you can only make 80, because 08 is just 8. With the same card twice (4 and 4) you can only make 44. So you can make two different two-digit numbers only when the two cards are different and neither is 0.**

**Look for:** A child who tests several pairs, notices that the tens digit decides, and finds the two special cases (a 0 card, two matching cards). A reason using tens and ones is better than "because it is greater".
**Common mistakes:** Saying 08 is a number you can make.; Saying 4 and 4 makes two numbers.

## Talk and explain (minutes 70 to 76)

**Put the smallest card on the left to make the smallest number. Is this always right? (a) Try the cards 3 and 8. Circle Yes or No. (b) Try the cards 0 and 5. Circle Yes or No. I know because ___.** Sentence starter: I think ... because ...

**Model answer: (a) Yes: 38 is smaller than 83. (b) No: the 0 can not go on the left, because 05 is just 5. The only two-digit number is 50. So the smallest card goes on the left, unless it is 0.**

**Look for:** A reason that uses tens and ones (the card on the left counts the tens), and a child who notices that 0 can not lead.
Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

## Exit ticket (minutes 76 to 82): 3 questions

**E1** **Write the greater number: 39 or 41.**

The greater number is ___

**Answer: 41**

**Worked solution:** 41 has 4 tens and 39 has only 3 tens, so 41 is greater. Look at the tens first.
**Common mistakes:** Writing 39 because 9 is greater than 1.

**E2** **Make two numbers with both cards 7 and 3.**

(a) The greater number is ___

(b) The smaller number is ___

> **Answer: 73 and 37**

**Worked solution:** The greater card on the left makes the greater number: 73. The smaller card on the left makes the smaller number: 37.

**Common mistakes:** Writing 73 for both.; Writing 10 (7 + 3).

**E3** **Arrange these in order. Begin with the smallest number: 63, 36, 60.**

___ , ___ , ___

> **Answer: 36, 60, 63**

**Worked solution:** 36 has 3 tens. 60 and 63 both have 6 tens. 60 has 0 ones and 63 has 3 ones, so 60 comes before 63.

**Common mistakes:** Putting 60 before 36 or after 63.; Writing them greatest first.

### Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (knowing which card goes on the left; looking at the tens first; the 0 card; listing every number (Think harder); something else); *What helped me* (digit cards or my kit; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (using my cards; building it with the kit; checking my answer; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

## 5. Home practice and online practice

| # | Tier | Question | Answer |
|---|---|---|---|
| H1 | Start here | You have the digit cards 4 and 7. Write the two numbers you can make. Circle the greater number. | 47 and 74. 74 is greater. |
| H2 | Start here | Arrange these in order. Begin with the smallest number: 72, 27, 70, 17. | 17, 27, 70, 72 |
| H3 | Start here | Ben says 38 is greater than 41. Is Ben right? Circle Yes or No. I know because ___. | No. 41 is greater, because 4 tens is more than 3 tens. |
| H4 | Think harder | You have the cards 0, 6 and 9. Use two cards each time. The smallest two-digit number is ___. The greatest is ___. | 60 and 96 |
| H5 | Go further | At home: make digit cards 1, 2, 3, 4 and 5. How many different two-digit numbers can you make with two different cards? Arrange them in order. Begin with the smallest. | 20 numbers: 12, 13, 14, 15, 21, 23, 24, 25, 31, 32, 34, 35, 41, 42, 43, 45, 51, 52, 53, 54 |

Online practice (`out/l03/practice`, key `t1l03`): 14 items; 8 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

## 6. Parent line for the term report

Your child is learning to see numbers as tens and ones and to say why. This week: making two-digit numbers from digit cards and comparing them, tens first.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

## 7. Slide list

| # | Time | Slide | On screen | Speaker note |
|---|---|---|---|---|
| 1 | 0–3 | **Digit Cards**<br>s-title | TODAY WE WILL: / Make numbers with digit cards, and say which is greater. | Lesson 3 of Term 1, 90 minutes. Objective: make two-digit numbers from digit cards and compare them, tens first. Time plan: opening 0 to 3, warm-up 3 to 9, sneak-peek 9 to 13, explore 13 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Say the lesson is recorded (the tutor, the shared screen and any unmuted voice) and a child may keep the camera off. |
| 2 | 0–3 | **Get ready**<br>s-cards | 0 1 2 3 4 5 6 7 8 9 / Your digit cards, 0 to 9, face up. / Your tens bars and ones squares. / A pencil and your workbook. | Show this while children arrive. Each child needs digit cards 0 to 9 (cut from the kit page). A child whose cards are not cut out writes the digits on ten scraps of paper now (about a minute). Check cameras show the cards and hands. Nothing is bought: scrap paper is enough. |
| 3 | 3–9 | **Warm-up: count in tens**<br>s-rem1 | Clap on every number / 10 20 30 40 50 60 70 80 90 100 / Now count back from 100. Whisper, and shout on 50! | Count in tens to 100 with a clap on each number, then back to 0. It is a spiral review of skip counting in tens (holiday topic) and it lets a new child join: they hear it before they say it. |
| 4 | 3–9 | **Tens and ones**<br>s-rem2 | [picture: bundles 3 tens 4 ones] / A DIGIT: / One of the symbols 0 1 2 3 4 5 6 7 8 9. / In 34, the left digit counts the tens. The right digit counts the ones. / 25 40 17 63 | Teach digit, tens and ones with the kit: 3 bars and 4 squares make 34. Then say each number on the slide and children say how many tens and how many ones (25 is 2 tens and 5 ones, 40 is 4 tens and 0 ones, 17 is 1 ten and 7 ones, 63 is 6 tens and 3 ones). Show with fingers or kit. No writing, no marks. |
| 5 | 3–9 | **More or fewer?**<br>s-rem3 | [picture: bundles 1 tens 3 ones] / [picture: bundles 0 tens 9 ones] / Which group has more squares? | Group A is a full bar of ten and 3 more squares, which is 13. Group B is 9 squares. Group A has more. Then ask by voice: 16 or 19 (19), 20 or 12 (20). Children point or answer with fingers. This is comparing two sets by matching and counting, to 20. Tens first comes in the mini-lesson. |
| 6 | 9–13 | **Sneak-peek puzzle 1**<br>s-sneak | Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Count the stars in each box. Which box has more? Write how many. / Box A / [picture: scatter] / Box B / [picture: scatter] | Puzzle 1 (box A has 15 stars, box B has 18) is an access question: any child can try it by counting or drawing rings. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Answer: box B, 18 stars. Do not reveal it until Talk time. Do not write scores or keep a list of who found it easy. |
| 7 | 9–13 | **Sneak-peek puzzle 2**<br>s-sneak2 | PUZZLE 2: think: / Ali has the digit cards 4 and 8. / He puts the 4 first and makes 48. / Mei swaps the cards and makes a new number. / Who has the greater number? Circle Ali or Mei. I know because ___. / 4 8 | Puzzle 2 is a thinking question on the key idea. Children write or draw on their own paper. Nobody holds paper to the camera: children keep their papers and check their own thinking at Talk time. There is no private chat and no board in front of the group. Answer: Mei, because 84 is 8 tens and 48 is only 4 tens. Do not reveal puzzle 2 until Talk time. Do not read anyone's answer aloud. |
| 8 | 13–28 | **Explore: two cards, two numbers**<br>s-explore1 | Put your digit cards face up. / Pick two cards. Put them side by side. / Read your number. Say the tens first. / Swap the cards. Read the new number. / WHAT IF?: / What if one of your cards is 0? | Round 1, about 4 minutes. Objects first, no writing. Children make three pairs, swap, and say both numbers aloud, tens first. Children keep their cards on the table and say both numbers aloud; nobody holds anything up to the camera. The what-if card is open from the start so nobody waits; it previews the zero rule and nobody is named for choosing it. Expect some children to read 47 as seven-four: ask them to say the tens first. |

| | | | | |
|---|---|---|---|---|
| 9 | 13–28 | **Build both numbers**<br>s-explore2 | Pick two cards that add up to 9 or less. Build both of your numbers with bars and squares. Which pile has more? / [picture: bundles 5 tens 3 ones] / [picture: bundles 3 tens 5 ones] | Round 2, about 4 minutes, main room. Children pick two cards whose digits add up to 9 or less (the kit has 9 bars), then build both numbers with the kit bars (tens) and squares (ones); a child may build one, read it, then build the other. The picture shows 53 and 35 as an example. They see that the pile with more bars is greater, so the tens decide. Ask: which pile is greater, and how do you know? |
| 10 | 13–28 | **Show me!**<br>s-explore3 | Make it, give a thumbs-up, then check your own. / Cards \| Make the / 7 and 2 \| greatest number / 3 and 9 \| greatest number / 0 and 5 \| greatest number / 6 and 1 \| smallest number | Round 3, about 4 minutes. Say each pair aloud and keep this table on screen. Children make the number and give a thumbs-up, then check their own against your line; nobody holds anything up to the camera. Answers: 7 and 2 make 72; 3 and 9 make 93; 0 and 5 make 50 (05 is just 5, settled on the next slide); 6 and 1 make 16. Leave the zero case open and let the children argue it. |
| 11 | 13–28 | **What did we notice?**<br>s-explore4 | Which card goes on the left for the greatest number? / Which card goes on the left for the smallest number? / What happens with a 0 card? Could you make 05? | Take one or two answers from volunteers. Key ideas: the greatest card goes on the left for the greatest number; the smallest card goes on the left for the smallest number; a number does not start with 0, so with 0 and 5 the only two-digit number is 50. Repeat a child's answer in your own words so the recording picks it up. |
| 12 | 28–38 | **Two cards, two numbers**<br>s-m1 | [picture: bundles 3 tens 5 ones] / [picture: bundles 5 tens 3 ones] / 3 5 / Same cards. The left card counts the tens. | Put the 3 first: 3 bars and 5 squares make 35. Swap the cards: 5 bars and 3 squares make 53. The same two cards make two different numbers because the left card counts the tens. Ask: which card is the tens card in 53? Answer: the 5. |
| 13 | 28–38 | **Which is greater? Tens first**<br>s-m2 | [picture: bundles 6 tens 1 ones] / [picture: bundles 5 tens 9 ones] / THE RULE: / Look at the tens first. / Same tens? Then look at the ones. | Think aloud: 6 bars is more than 5 bars, so 61 is greater than 59, even though the ones digit 9 is greater than 1. Many children look at the ones first. Then a same-tens pair: 47 and 43 have 4 tens each, so look at the ones: 7 is more than 3. Ask: which is greater, 52 or 58? Answer: 58. |
| 14 | 28–38 | **Your turn: which is greater?**<br>s-m3 | Write the greater number each time. Timer: 1 minute / 28 or 82 / 61 or 59 / 34 or 38 | Each child writes three numbers on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. Children may build with the kit. |
| 15 | 28–38 | **Check: which is greater?**<br>s-m3a | 28 or 82 / 82 / 61 or 59 / 61 / 34 or 38 / 38 | Answers: 82, 61, 38. Read each answer aloud so a child watching the recording can mark their own work. For 34 and 38 the tens are the same, so look at the ones. |
| 16 | 28–38 | **Greatest and smallest**<br>s-m4 | 7 2 / Greatest: greatest card on the left / 72 / Smallest: smallest card on the left / 27 | Worked example with the cards 7 and 2: the greatest number has the greatest card on the left (72); the smallest number has the smallest card on the left (27). Then ask for the cards 4 and 9 (94 and 49) without showing them. Both numbers use the same two cards. |
| 17 | 28–38 | **The 0 card**<br>s-m5 | 0 6 / 6 then 0 is 60. A two-digit number. / 0 then 6 is 06. That is just 6. / A number does not start with 0. | The zero rule in one minute. With the cards 0 and 6 the only two-digit number is 60. 06 is just 6: a zero on the left is empty. So when we make the smallest two-digit number and one card is 0, the 0 goes on the right. Ask: why is 06 not a two-digit number? Do not use the word place value. |
| 18 | 28–38 | **Put them in order**<br>s-m6 | Smallest first. Tens first! / 45 54 39 40 | Smallest first: 39, 40, 45, 54. Think aloud: 3 tens is the fewest, so 39 is first. 40 has 4 tens and 0 ones, 45 has 4 tens and 5 ones, so 40 comes before 45. 54 has 5 tens, so it is last. Children may say the order or write it. |

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| 19 | 38–42 | **Tall and small**<br>s-break | First number greater? Reach up tall. / First number smaller? Curl down small. / The same? Hands on hips. / NUMBER PAIRS: / 52 and 25 / 38 and 83 / 61 and 59 / 40 and 40 / 19 and 21 / Sip some water. | Four minutes. Stand up on camera. Call each pair of numbers: children reach up tall if the first is greater, curl small if the first is smaller, hands on hips if they are the same. Pause after each pair and ask one child to say how they know. Keep it gentle for small spaces. No equipment. |
| 20 | 42–70 | **Workbook page 1: Start here**<br>s-ws1 | Questions 1 to 3. Timer: 8 minutes / questions S1, S2, S3 | Read questions 1 and 2 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. |
| 21 | 42–70 | **Check page 1**<br>s-ws1ans | answers S1, S2, S3 | Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group. |
| 22 | 42–70 | **Workbook page 2: Start here**<br>s-ws2 | Questions 4 to 8. Timer: 10 minutes / questions S4, S5, S6, S7, S8 | Read question 6 aloud: it is a spot-the-mistake. Read questions 7 and 8 aloud as well. Children may build the numbers with the kit to decide. |
| 23 | 42–70 | **Check page 2**<br>s-ws2ans | answers S4, S5, S6, S7, S8 | Read each answer aloud so a child watching the recording can mark their own work. Question 6 answer: no, 21 is greater, because 2 tens is more than 1 ten. Question 8 answer: Ravi, 5 tens is more than 2 tens. Never ask a child to show a wrong answer to the group. |
| 24 | 42–70 | **Choose your next page**<br>s-choice | Page \| What it is / Think harder (page 3) \| Our idea in a new setting / Go further (page 4) \| Harder puzzles / Investigation (page 5) \| A pattern to find / Check \| Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too. | Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes. |
| 25 | 70–76 | **Talk and explain**<br>s-talk | TALK ABOUT IT: / Put the smallest card on the left to make the smallest number. / Is this always right? (a) Try the cards 3 and 8. (b) Try the cards 0 and 5. / I think … because … | One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: for 3 and 8 it works: 38 is the smallest. For 0 and 5 it does not: 0 can not go on the left because 05 is just 5, so the only two-digit number is 50. Repeat each child's answer in your own words for the recording. |
| 26 | 70–76 | **Back to the sneak-peek**<br>s-sneakans | [picture: bundles 4 tens 8 ones] / [picture: bundles 8 tens 4 ones] / Puzzle 1: box B, 18 stars. Puzzle 2: Mei has the greater number, 84. | Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone's earlier answer. Puzzle 1: box A has 15 stars, box B has 18. Puzzle 2: 84 is 8 tens and 4 ones, 48 is 4 tens and 8 ones, so Mei has more. |
| 27 | 76–82 | **Exit ticket**<br>s-exit | On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3 | Children answer on page 5. Read E1, E2 and E3 aloud (the questions are on the slide). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat. |
| 28 | 76–82 | **Check the exit ticket**<br>s-exitans | answers E1, E2, E3 | Read each answer aloud. Children check their own ticket in a different colour. |

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| 29 | 82–85 | **Look-back**<br>s-look | [look-back box] | Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional. |
| 30 | 85–90 | **Well done, inquirers!**<br>s-end | Show a grown-up one thing you found out today. / Next lesson: The 100 Square Explorer. / Have ready: 100 square, 12 counters, colours. | Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide. |
| 31 | optional | **Go further: what if?**<br>s-opt | How many two-digit numbers have two matching digits, like 33? / Write them all. What do you notice? / 33 44 | Optional, only if you are on time. Answer: 11, 22, 33, 44, 55, 66, 77, 88, 99. That is nine numbers. When you swap the digits you get the same number. Take two answers and move on. Do not teach adding 11. |
| 32 | optional | **If we have time: pick one**<br>s-menu | Activity \| What we do / Is it always true? \| Is the greater card on the left always the greater number? Test it. / Make a puzzle \| Write a digit card puzzle for your grown-up / Teach it back \| Show how to tell 61 is greater than 59 / Optional. Everyone may join. | Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum. |

## 8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

| Chapter | Part | You say (recording voice) | A child watching later |
|---|---|---|---|
| 0:00 | Opening | *If you are watching the recording: hello! Make your digit cards 0 to 9, get your workbook and a pencil. Pause or rewind any time.* | Make digit cards 0 to 9 on scrap paper. Gather your workbook, kit and pencil. Press play when ready. |
| 3:00 | Remember warm-up | *Recording: join in! Say the numbers out loud and show tens and ones with your bars and squares. Pause if you need longer.* | Play along. Say and show each answer before the video does. |
| 9:00 | Sneak-peek puzzles | *Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.* | Pause. Try both puzzles on your own paper. A guess is fine. |
| 13:00 | Explore | *Recording: pause now. Pick two cards, make both numbers, then build them with bars and squares. Press play when you have tried.* | Pause. Make and swap pairs of cards, build both numbers, then try the cards I call. |
| 28:00 | Mini-lesson | *Recording: keep watching and copy the examples. Rewind if you miss a step.* | Watch and copy the examples. Rewind any step you missed. |
| 38:00 | Brain break | *Recording: stand up and join in! Reach up tall for the greater number, curl small for the smaller one.* | Stand up and show tall or small for each pair of numbers. |
| 42:00 | Workbook | *Recording: pause now and do workbook page 1. When you finish a page, press play to check it.* | Pause. Do page 1, then play to check. Then page 2. Then choose a page to try. |
| 70:00 | Talk time | *Recording: pause. Tell a grown-up your answer to the talk question, and say why.* | Pause. Explain your answer to someone at home, with a reason. |

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| 76:00 | Exit ticket | *Recording: pause and do the exit ticket on your own. Then press play to check it.* | Pause. Do the exit ticket alone, then play to check it. |
| 82:00 | Look-back | *Recording: pause and fill in your look-back box. You can pass.* | Pause. Fill in the look-back box, or pass. |
| 85:00 | Flex time and goodbye | *Recording: well done for finishing! Show a grown-up one thing you found out.* | Show a grown-up your workbook and tell them one thing you found out. |

## 9. Timing check (model, not a stopwatch)

| Part | Allocated | Script | Slow class | Flag |
|---|---|---|---|---|
| opening | 3:00 | 2:38 | 3:02 | OVER (slow class) |
| remember | 6:00 | 4:54 | 5:46 | ok |
| sneak | 4:00 | 3:29 | 4:05 | OVER (slow class) |
| explore | 15:00 | 13:29 | 15:58 | OVER (slow class) |
| mini | 10:00 | 8:07 | 9:34 | ok |
| break | 4:00 | 3:24 | 4:01 | OVER (slow class) |
| work | 28:00 | 26:53 | 32:05 | OVER (slow class) |
| talk | 6:00 | 5:30 | 6:32 | OVER (slow class) |
| exit | 6:00 | 4:07 | 4:52 | LOOSE |
| look | 3:00 | 2:43 | 3:12 | OVER (slow class) |
| flex | 5:00 | 0:44 | 0:50 | ok |
| **Total** | 90:00 | 75:58 | 89:57 | |

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 8 questions; a middle child needs about 7.0 min for page 1 and 10.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.